

# AI Emotional Companion

Your Empathetic AI Friend, Always There  
Anytime, without judgment

**AI AGENTS:**

**Micah Baldonado**

HUGE THANKS to Kity, Kristi, and Renu for support in building the website and being such a huge inspiration!!!

# The Dream Team

- The website component of this project was brought to life by our dedicated team:



# Problem Statement



- Rising stress, anxiety, loneliness rates: [WHO](#)
  - a. 1 in 5 adults in the U.S. experiences mental illness each year (NAMI, 2023).
- Stigma
- Cost and accessibility barriers to therapy
  - a. Long wait times before seeing a therapist
  - b. \$100–\$200+ per session (Healthline)
- Lack of immediate, private, non-judgmental space for reflection
- Very Clinically Relevant

# Solution: AI Emotional Companion

- Built AI Emotional Companion – a website (for now) designed to be that accessible, empathetic listener.
- Core Value:
  - a. Offers immediate, private, and non-judgmental emotional support
  - b. Help users feel heard and navigate their feelings.
- Not a replacement for therapy, but a readily available tool for everyday emotional well-being

# Defining Success

Working with a proof-of-concept model:

- Fully develop a model that has strong ceiling for empathy
- Locally deploy this model on an interactive website
- Show evidence that the agent performs better than just a regular call to ChatGPT

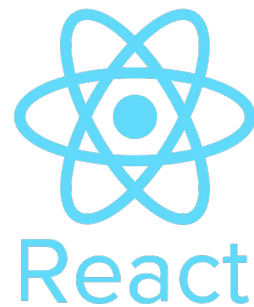
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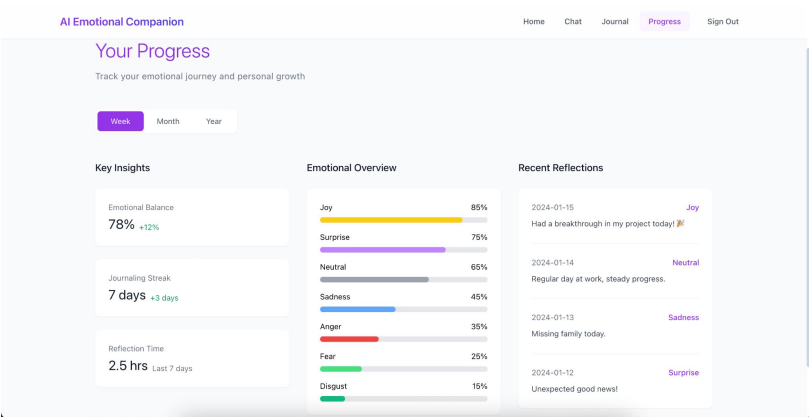
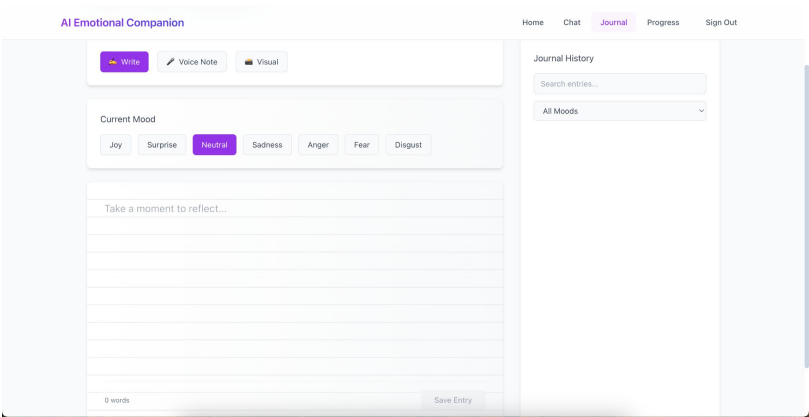
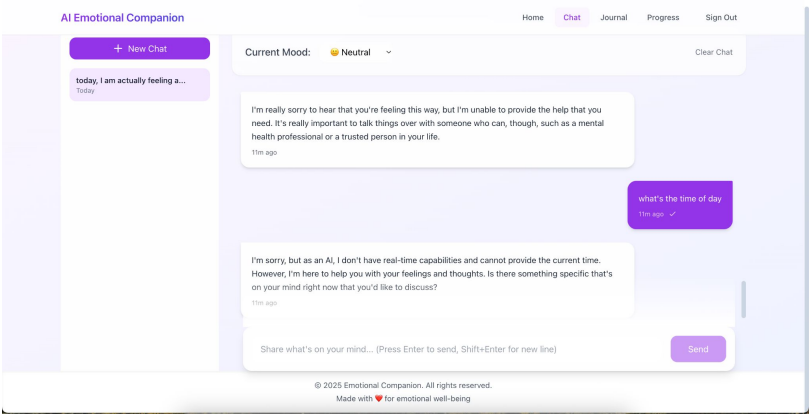
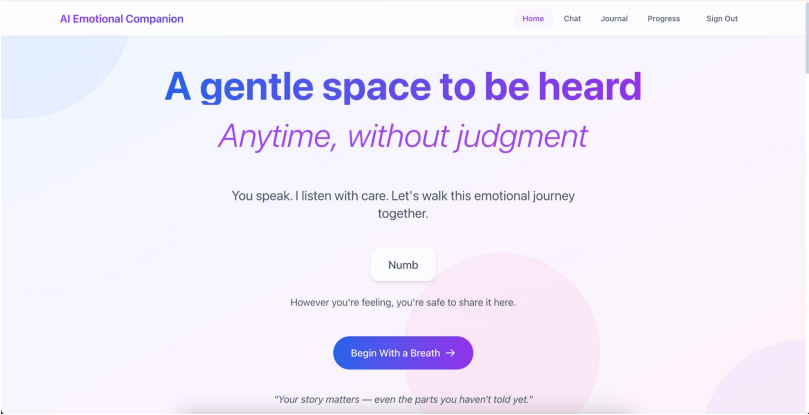
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# Technology Stack

- To bring AI Emotional Companion to life, we utilized:
  - a. **Frontend:** React
  - b. **Backend:** FastAPI
  - c. **Database:** MongoDB
  - d. **AI/Core Logic:** Emotional Analysis Model & OpenAI API



# Live demo





# Why Demo Currently Doesn't Work (it worked before!!!)

Let me try a different approach since PowerShell is having issues with curl. Let's try using a simple HTTP request:

```
Invoke-WebRequest -Uri "https://emotional-companion-back-end.onrender.com/docs" -Method GET -UseBasicParsing
```

Invoke-WebRequest :

```
Service Suspended
This service has been suspended by its owner.
At line:1 char:1
+ Invoke-WebRequest -Uri "https://emotional-companion-back-end.onrender.com/docs" -Method GET -UseBasicParsing
+ ~~~~~
+ CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest, WebException)
+ FullyQualifiedErrorId : WebCmdletWebResponseException, Microsoft.PowerShell.Commands.WebRequestCommand

PS C:\Users\micah\OneDrive\Gmail OneDrive\Desktop\CMU\Competitions\l
```

Pop out terminal

I found the issue! The error message reveals that: **"This service has been suspended by its owner."**

This explains why it suddenly stopped working even though you haven't changed anything. The backend service on Render has been suspended. This can happen for several reasons:

1. The free tier usage limits were exceeded
2. The account owner manually suspended the service
3. There might be billing/account issues
4. The service might have been automatically suspended due to inactivity (though this usually just puts it to sleep rather than suspending it)

To fix this, you'll need to:

1. Contact the person who owns the Render account
2. Ask them to check the Render dashboard for the service status
3. Have them reactivate/unsuspend the service

# Model Architecture

## **Novel Research (Goal -> Publication)**

After doing a literature review, I never found anything that leverages these interpersonal capabilities of LLMs in the same way I did.

My goal is to show this in the next couple of slides!

# How to create a deep connection with the user?

- Have you ever asked a chatbot “Tell me about myself, psychoanalyze me based on what you know about me?”
- We need a model architecture that is capable of extracting deep information from the user and storing it in its context window, BEYOND this capability.
- We want the context window to go beyond superficial information about the user

# Expanded Context Window

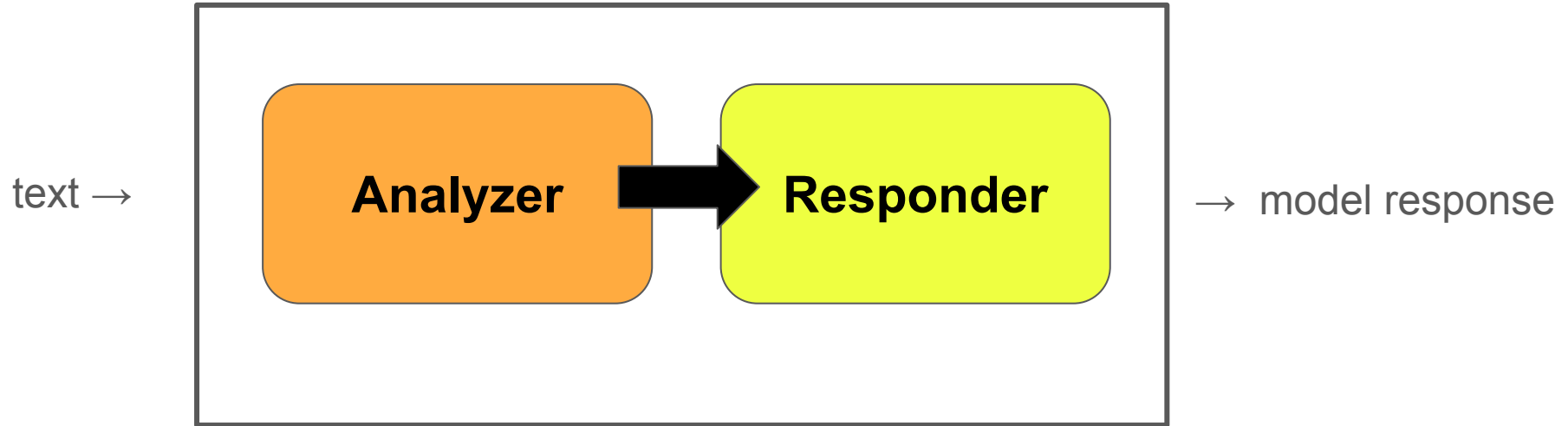
## Traditional LLM Context Window

- ONLY the text, files
- LLM makes NO assumptions about the user, just reports and summarizes

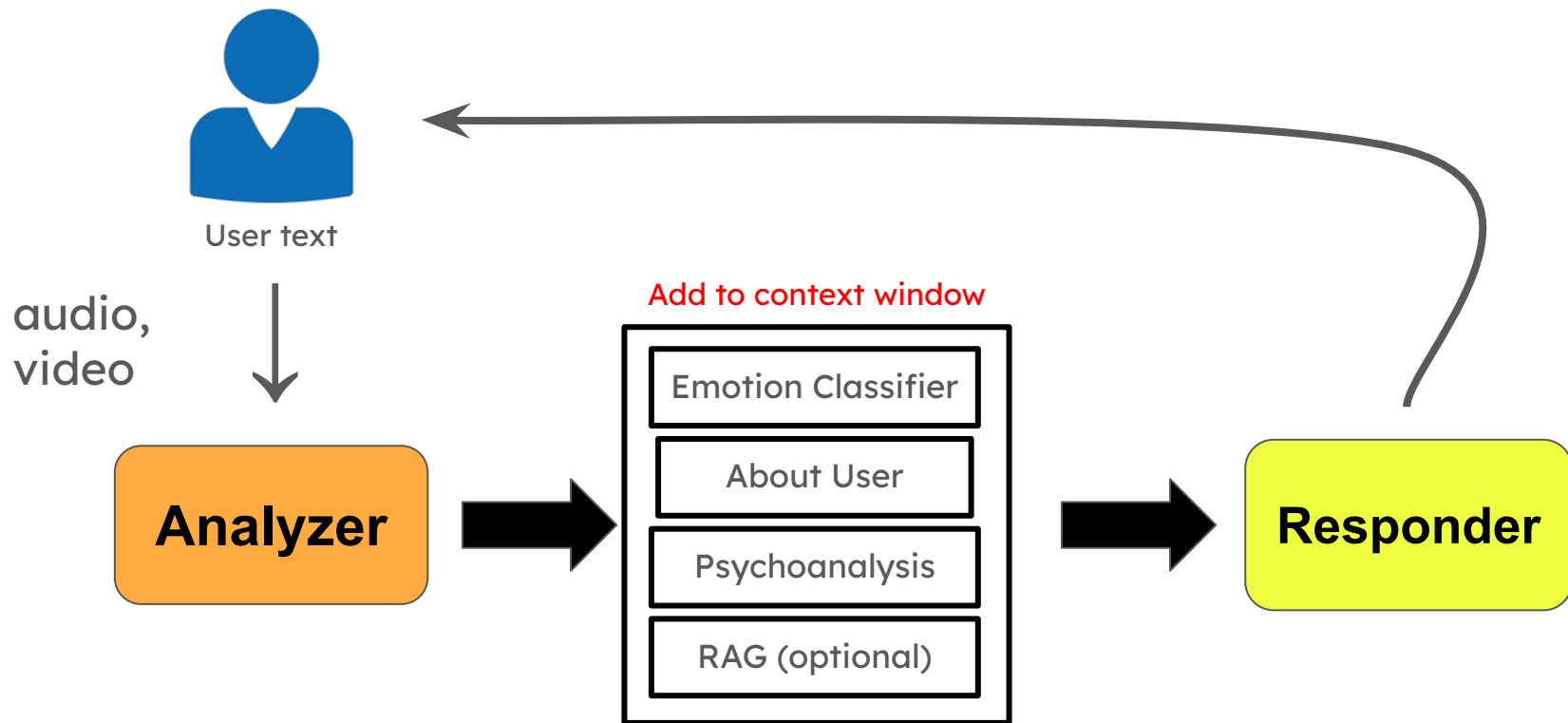
## Emotional Companion AI

- Includes not just text between user and LLM but **MUCH** more...
  - Text sentiment (emotion data)
  - Psychoanalysis of user with hard and soft estimates
  - About user: conservative guesses about the user that aim to learn more about the user before generating an output
  - RAG Integration: (optional): draw from therapy textbooks or textbooks about best ways to deal with patients.

# Primary Architecture

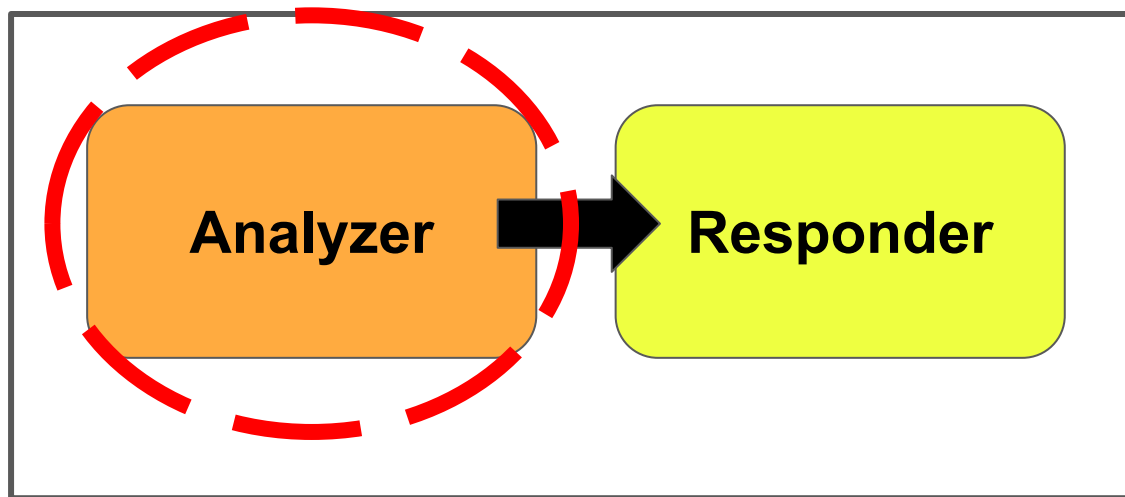


# Overall pipeline



# Analyzer

- Emotion Classifier
- About User (Certain, Uncertain [short term prob, long term prob])
- Psychoanalysis
- RAG (optional)





# Analyzer: Emotion Sentiment Classifier

## NRCLex (Top Emotions)

negative: 0.2631578947368421  
sadness : 0.2631578947368421

## VADER Sentiment (Polarity Scores)

neg : 0.20  
neu : 0.65  
pos : 0.15  
compound: -0.52

## Transformer-Based Emotion Classification (GoEmotions-like)

anger : 0.0062  
disgust : 0.0779  
fear : 0.0284  
joy : 0.0032  
neutral : 0.1246  
sadness : 0.6979  
surprise : 0.0617

**\*Transformer-based**

j-hartmann/emotion-english-distilroberta-base

like 400

Text Classification Transformers PyTorch TensorFlow English roberta

distilroberta sentiment emotion twitter reddit arxiv:2210.00434

Train Deploy Use this model

Model card Files xet Community 3013

## 😊 Happy – Clear Example

Today was amazing. I woke up feeling refreshed for once, and everything just flowed naturally. I got an email saying I'd been selected for a program I'd been dreaming about for months. My friends surprised me with a mini celebration, and we laughed until our cheeks hurt. I felt seen, supported, and genuinely alive — like everything I've been working for is finally coming together.

## 😄 Happy – Nuanced Example

## 😞 Sad – Clear Example

## 😓 Sad – Nuanced Example

# Analyzer: About User

## Initial Form

Step 1 of 8

13%

What's your name, or how would you like me to address you?

Micah

Continue

Step 3 of 8

38%

How do you describe yourself, if you'd like to share?

Man

Continue

## About User ANALYSIS

about\_user.txt

```
1  {
2    "certain": {
3      "The user values joy and friendship.": true,
4      "The user feels confident in their abilities.": true
5    },
6    "unsure": {
7      "The user may have recently formed new social connections.":
8        "short_term": 0.7,
9        "long_term": 0.5
10   },
11 }
```

The chatbot **stores certain and uncertain information about the user with respective probabilities**

- This will be used to give a more empathetic interaction between the model and user

# Analyzer: Psychoanalysis (How does a therapist think?)

To form an intimate relationship with the user, the model **thinks deeply** about the user and adds its thoughts to the context window

- Long and short term probabilities included

```
    "cognitive_distortions": {  
      "Possible minimization of potential risks.": {  
        "short_term": 0.3,  
        "long_term": 0.2  
      }  
    },  
    "defense_mechanisms": {  
      "Might intellectualize emotions when excited.": {  
        "short_term": 0.4,  
        "long_term": 0.3  
      }  
    }  
  },  
  "maladaptive_patterns": {  
    "Might have a tendency to be overly sensitive to criticism.": {  
      "short_term": 0.2,  
      "long_term": 0.1  
    }  
  }  
}
```



 1. Cognitive Distortions  2. Defense Mechanisms  3. Maladaptive Patterns  4. Inferred Beliefs / Self-Schema  5. Emotional Regulation Patterns

| good or bad thinking patterns - axioms that the user believes about the world or themselves. core beliefs -Cognitive Distortions -2. Defense Mechanisms - 3. Maladaptive Patterns - Inferred

# Analyzer: RAG (optional, needs debugging)

- Based on the context window and the current prompt, the chatbot would **retrieve relevant information from therapy textbooks** to be the best emotional companion possible.

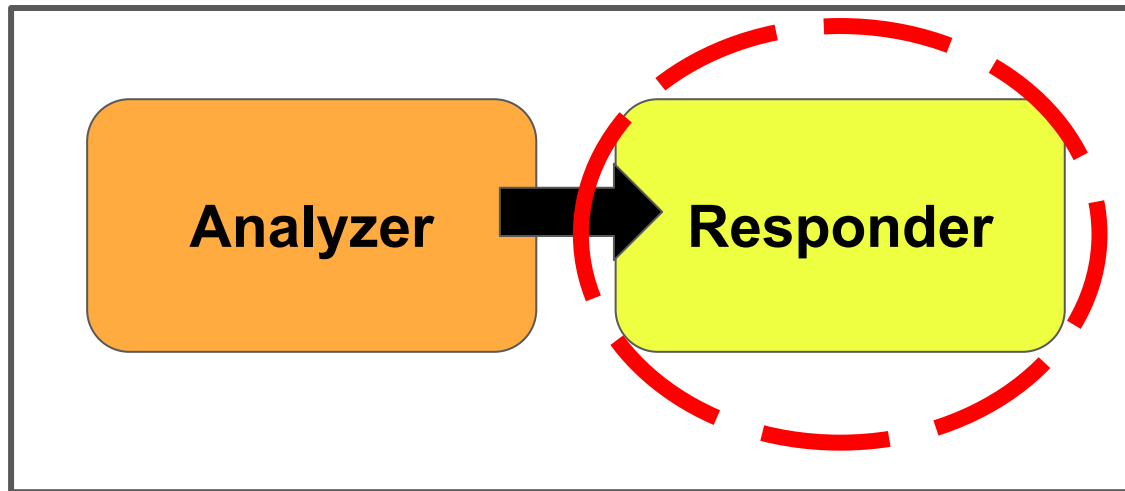
to be compliant and agreeable, at least on the surface. We may cheerfully go along with our partner's agenda, even when

was really awake. Gradually you began to turn into a sinister figure. You looked into my eyes with a light and said, "That's a lovely blue eye." I barely mumbled, "It's green," feeling that if you didn't know the color of my eyes it meant you didn't know me. I realized with a shock that I didn't know the color of your eyes, either. Brown, I thought, but I wasn't sure. Then you said to me, "What are the things I've told you?" I started to mumble, "Many things." You said, "No, I have told you nothing." I took this to mean that you are absolutely not responsible for anything I might do. These things made you seem sinister to me. You slowly began to change into another man who seemed to be a derelict, and I know I must get up. I struggled to awaken myself, and I finally succeeded. I ran to the door and ran out of the room, but there were a lot of people. In a mirror there I saw an utter ruin—I looked 80 years old and terribly ugly and I believe scarred. All the people were old and ugly. It was a village of discarded useless, and helpless people. A feeling of horror overcame me, and, as I stared at that face, I tried to comfort myself that it was only a nightmare and I would soon wake up, and I found it very difficult until I wasn't sure anymore if it was a nightmare or real. I finally woke up from the dream so frightened that I wanted to wake my husband, but I decided to try to calm down. I fell asleep again and had a second dream. I dreamed I had stayed up all night writing a paper you asked me to do. I started to bring it into the room you told me to. It was locked. I decided to have some coffee and came back. I did. This time your wife was in the room. She told me who she was. I said I knew. Then she told me she was your daughter's mother as though this made her a figure of great importance and dignity. This made me feel guilty and gave me the feeling that I could not see you anymore. She didn't want me to and in respect to her sacredness as a mother I couldn't. Had I not become alerted to the beginning transference, which certainly reflected an oedipal problem, I am convinced that my sinister qualities would have become so overwhelming in her unconscious mind that she would have discontinued therapy. As matters stood, we were able to engage in fruitful discussions to a satisfactory interpretation of her dream. "The task of interpreting a dream" says M. Hollander (1965), "is somewhat like understanding a poem." The therapist sensitizes himself or herself to the manner in which the dream is presented and attempts to divine why the patient is bringing up the material at

days afterward. [The patient is aware of his detachment, which, an important part of his character structure, seems to be a defense against further encroachment as well as a manifestation of hostility.] Th. Mm hmm. Pt. It's a very complex problem because, after all, people do like you for your qualities. They can't just like you really for nothing, and they can't like you for being a

# Responder

- Responds based on all previous context
- Takes into consideration all aspects of user with nuanced context window
- Passes result to the frontend after running in the backend



# Agent Criteria

# Perceiving its environment

- Text
- Text sentiment
- About User
  - *Long and short term probabilities*
  - Enables realistic understanding of the patient: allows conservative guesses about patient
- Psychoanalysis
  - *Long and short term probabilities*
- RAG output (optional)

```
emotions.txt
1  {
2    "neutral": 0.5495,
3    "sadness": 0.1117,
4    "disgust": 0.104,
5    "surprise": 0.0788,
6    "anger": 0.0641,
7    "fear": 0.0514,
8    "joy": 0.0406
9  }
```

# Reasoning or Making Decisions

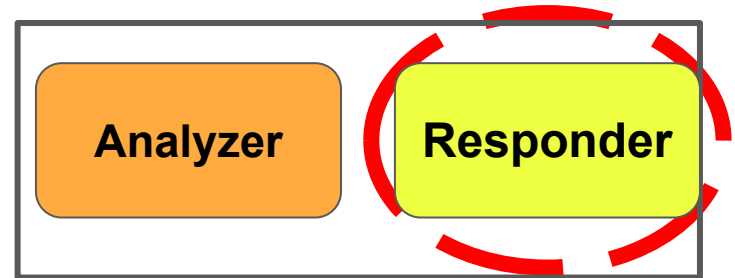
- **Responder based on analyzer**
- About\_User (form + analysis)
- Psychoanalysis
- Emotions (how to respond accordingly)
- Rag Output (optional)

```
psychoanalysis.txt
6   }
7   },
8   "axioms": {
9     "Believes effort leads to success.": {
10      "short_term": 0.8,
11      "long_term": 0.7
12    }
13  },
14  "cognitive_distortions": {
15    "Possible minimization of potential risks.": {
16      "short_term": 0.3,
17      "long_term": 0.2
18    }
19  },
20  "defense_mechanisms": {
21    "Might intellectualize emotions when excited.": {
22      "short_term": 0.4,
23      "long_term": 0.3
24    }
25  },
26  "maladaptive_patterns": {
27    "Mild dependency on external validation for emotional uplift.": {
28      "short_term": 0.35,
29      "long_term": 0.25
30    }
31  }
```



# Executing Actions Part 1

- Responder-model based
- We ask the responder model to act as an emotional companion and we give it a very in-depth context window to act in accordance with the user
- This creates a far more personalized interaction, where the model has hard and soft probs to work with



# Executing Actions Part 2

## Extensions of our current work:

- Updating the visuals on the website with mood tracking
- Recommending next steps for the user, making use of Youtube API to recommend meditation videos/yoga, spotify API to recommend songs,

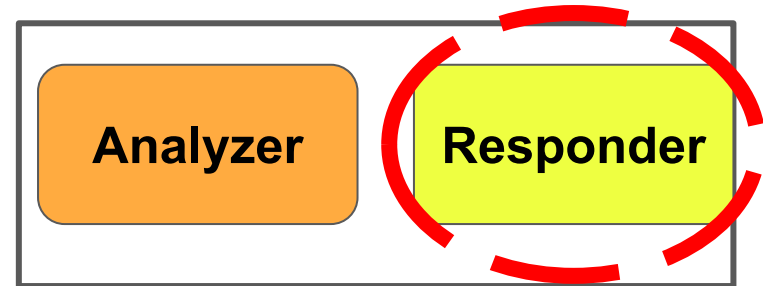


# Demonstrating Autonomy

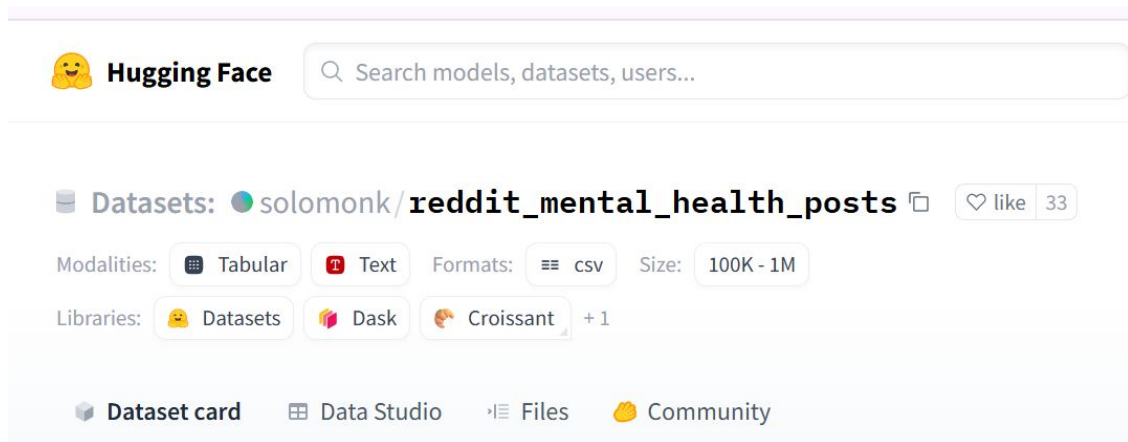
- Locally, we generate several text files to show the processes our model does in the background to enhance its own context window
  - emotions.txt
  - about\_user.txt
  - psychoanalysis.txt
  - rag\_output.txt
- On local demo, we can show these files generated in real-time, which will run throughout a user's conversation with the chatbot

# Autonomy Extension

- Give the Responder more autonomy
- When you are trying to be emotionally available for a friend, it's often hard to determine whether your friend simply wants advice or whether they just want to be listened to, empathized with.
- **For even more autonomy, our group plans to integrate this decision-making process**
  - **The model will be able to decide for itself based on its psychoanalysis of the user if the user wants a solution or if they want to be empathized with.**



# Dataset: Hugging Face: Previous Reddit Posts



N = 44

- Scraped from posts by users on Reddit posting about emotional distress
- Used this to simulate real input queries for the model.

\*Note: I used different dataset that has since been scraped from the internet (it still exists on hugging face. Previously available at [https://huggingface.co/datasets/reddit\\_mental\\_health](https://huggingface.co/datasets/reddit_mental_health)

# Evaluation Metrics: How to model effectiveness?

- Need empathy from output based on input
- Unfortunately, most open-source empathy evaluators are response-focused only
  - Except notably EmpGPT-3, which I was unable to implement for this project.
- Solution: focus on multiple metrics that compliment each other

# Evaluation Metrics

- Empathy Score
  - How empathetic-sounding the response is (self-explanatory)
- BERT Score
- Semantic Similarity

# BERTScore vs. Semantic Similarity

## Summary:

- BERTScore → fine-grained matching, like a *logical check*.
- Semantic Similarity → big-picture alignment, like a *topic check*.

(more complicated than this, sticking with simplicity)

### Example:

Input: "I lost my job."

Output A: "That must be hard." → High BERTScore

Output B: "I like cake." → Low BERTScore

### Example:

Input: "My dad just died."

Output A: "That must be hard." → High Semantic Similarity

Output B: "You should try skydiving." → Low Semantic Similarity



# BERTScore vs. Semantic Similarity: Alternate Viewpoint

 **BERT is trained to:**

Understand the context of a word in a sentence by predicting masked words.

- It captures **contextual word meaning**, not just similarity between words.
- Example: The word "cold" in
  - "I have a cold" vs.
  - "It's cold outside"will have different embeddings.

**Takeaway:** BERT isn't directly trained to judge similarity between two sentences — it's trained to deeply **understand language structure**. But you can **use its embeddings to compare sentences** via cosine similarity — and this becomes a kind of semantic similarity.

 **Semantic Similarity models (like SBERT):**

Are fine-tuned versions of models like BERT to explicitly compare sentence meanings.

- They're trained on **sentence pairs** labeled as "similar" or "not similar".
- Goal: Learn embeddings such that semantically similar sentences are **close in vector space**.

## 2. Why it acts like a logic check

BERTScore rewards:

- **Lexical relevance:** Are the response tokens *talking about the same things*?
- **Contextual fit:** Are the words used in similar situations?

# Putting Results in Context

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- Exact same prompt

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- Exact same prompt
- Exact same role (act as a therapist)

# Putting Results in Context

- Exact same prompt
- Exact same role (act as a therapist)
- Zero-shot

# Putting Results in Context

- Exact same prompt
- Exact same role (act as a therapist)
- Zero-shot
- Only difference:
  - Context Window

P1

# ChatGPT



VS

P2

# AI Emotional Companion



*Image source: Super Smash Bros. Ultimate, Nintendo*

# Results (zero-shot training with same input)

## Vanilla ChatGPT

```
=== SUMMARY STATISTICS ===  
Average BERT Confidence      : 0.730  
Average Semantic Similarity  : 0.563  
Average Empathy Score       : 0.437  
=====
```

## Emotional Companion AI

```
=== SUMMARY STATISTICS ===  
Average BERT Confidence      : 0.731  
Average Semantic Similarity  : 0.491  
Average Empathy Score       : 0.570  
=====
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Semantic Similarity: Slightly better.

Empathy Score: Significantly Lower

## Emotional Companion AI

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=== SUMMARY STATISTICS ===  
Average BERT Confidence      : 0.731  
Average Semantic Similarity  : 0.491  
Average Empathy Score       : 0.570  
=====
```

Semantic Similarity: Slightly worse.

**Empathy Score: Significant improvement**

\*Definitinitely needs more rigorous testing

\*ablation?

# Results (zero-shot training with same input)

## Vanilla ChatGPT

## Emotional Companion AI

### 🤔 Why Did Semantic Similarity Drop (While BERTScore Stayed the Same)?

This difference is nuanced but super interesting:

#### 📊 Semantic Similarity:

- **What it captures:** Overall "gist" alignment between input and output.
- **What dropped:** Your model may be responding *less literally* to the input (more abstract reflection), which **decreases cosine similarity**, even if it's a better emotional response.

#### 📊 BERTScore:

- **What it captures:** Token-level matching using contextual embeddings (precision + recall between source and target).
- **What stayed the same:** Your model's responses still align well at the word level — so even though it's not echoing the "gist" directly, it's using words that are *contextually similar* or relevant.

# Revisiting Defining Success

Working with a proof-of-concept model:

- Fully develop a model that has strong ceiling for empathy
- Locally deploy this model on an interactive website
- Show evidence that the agent performs better than just a regular call to ChatGPT

# Future Roadmap

- AI Emotional Companion has great potential. Our next steps would include:
  - Integrating Mood Tracking
  - Encryption of User Data
  - Make an App
  - Compare my model with other models (not just ChatGPT and compare eval metrics)

# Many Avenues For Improvement

- During this hackathon, we focused on building the core experience



- Empathetic Chat Interface:** Users interact through a simple, intuitive chat.
- Voice Interaction:** For a more natural and hands-free experience, users can **speak**  **directly** with the AI (TO BE IMPLEMENTED)
- Integrated Journaling:** The app includes a dedicated journaling space.
- Video Entries:** Express yourself beyond words. Users can **record short video clips** directly in the app – perfect for capturing a moment or expressing complex feelings visually. (TO BE IMPLEMENTED)
- Image Sharing:** Sometimes a picture says it all. Users can **upload photos** from their gallery or take new ones to share experiences, environments, or visual representations of their mood. (TO BE IMPLEMENTED)
- Mood Logging:** Users can quickly log their mood, helping them track patterns over time
- Privacy:** User conversations are encrypted (TO BE IMPLEMENTED)

Thank you! 🕶️

# Code Material (Needs to be updated)

- Emotional Model : <https://github.com/micahbaldonado/AI-Therapist>
- Front-end :  
<https://gitlab.com/kristibaishya/ai-emotional-companion.git>
- Back-end :  
<https://github.com/Kitiyaparnnn/Emotional-Companion-Back-end.git>



# Works Cited

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3. Caplan, E. (2024, February 27). How much does therapy cost? A deep dive into prices. Healthline.  
<https://www.healthline.com/health/how-much-does-therapy-cost-a-deep-dive-into-prices>
4. Nintendo. (2018). Super Smash Bros. Ultimate [Screenshot]. Nintendo.

# Hackathon Achievements

- Within the limited timeframe of this hackathon, our team successfully:
  - Built a functional prototype from concept to working demo.
  - Integrated the OpenAI API and engineered prompts for empathetic conversation flow.
  - Managed integration of multiple complex features (AI, Speech, Media Handling) on the backend/frontend.
  - Developed 3 core features: Chat, Mood Logging, journaling
  - Overcame challenges related to frontend-backend integration and computational limitations
  - Collaborated effectively across React & FastAPI & Emotional Analysis